No.	Status	Date
LNDE-RPT-20191001-027-DS-4-RA-104	Issued	2019-10-18

UT Thickness Data Sheet - 4-RA-104

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-104, located within the ATU-100. The scope of work involves the recording of wall thickness at designated condition monitoring locations to establish a re-baseline for asset integrity management.

Trevor Nolan performed the ultrasonic thickness measurements and data recording. All activities were conducted in accordance with the requirements specified in QSCP-NDT-001 Rev 2 and the guidelines of ASTM E797.

Component Specifications

The subject line, 4-RA-104, is the rich side of the L/R exchanger. The piping is constructed from A106 Gr.B with a nominal outside diameter of 114.3 mm and a nominal wall thickness of 8.56 mm.

The design parameters for this line are 862 kPa(g) and 127 °C. The minimum allowable thickness (t-min) is defined as 3.81 mm.

Inspection Methodology

Thickness data was acquired as part of the re-baseline survey detailed in LNDE-RPT-20191001-027-R0. The measurements were executed following the site NDE procedure for ultrasonic thickness measurement in the ATU-100.

The survey focused on 150 CMLs across the unit to determine current wall loss and calculate remaining life based on the established retirement thickness of 3.81 mm.

Thickness Measurement Results

Measurements at the following CMLs were recorded: 2-RF-102-04 at 2.83 mm, 2-RF-102-05 at 2.88 mm, 2-RF-102-06 at 2.74 mm, and 2-RF-102-07 at 2.82 mm.

All four recorded values are below the t-min of 3.81 mm. These findings necessitate the issuance of Non-Conformance Reports (NCR) per site rules.

Remaining Life and Analysis

Based on the current readings and calculated corrosion rates, the remaining life for the measured points is 4.5 years for 2-RF-102-04, 4.3 years for 2-RF-102-05, 2.9 years for 2-RF-102-06, and 3.2 years for 2-RF-102-07.

The remaining life calculations are used to determine the next inspection interval, which must not exceed half of the remaining life or the class maximum for rich amine lines.

Thickness readings, measured 2019-10-18

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2016-02-12	6.17	5.93	0.050	0.065	36.3	2022-10-18	
6-SG-101-02	7.11	3.56	2016-02-12	5.76	5.51	0.065	0.068	28.7	2022-10-18	
6-SG-101-03	7.11	3.56	2016-02-12	5.87	5.56	0.064	0.084	23.7	2022-10-18	
6-SG-101-04	7.11	3.56	2016-02-12	5.58	5.34	0.074	0.065	24.0	2022-10-18	
6-SG-101-05	7.11	3.56	2016-02-12	5.38	5.04	0.089	0.092	16.0	2022-10-18	
6-SG-101-06	7.11	3.56	2016-02-12	5.88	5.56	0.064	0.087	23.0	2022-10-18	
6-SG-101-07	7.11	3.56	—	—	6.24	—	—	—	2022-10-18	
6-SG-101-08	7.11	3.56	2016-02-12	6.00	5.77	0.065	0.063	34.2	2022-10-18	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2016-02-12	5.42	5.05	0.089	0.101	14.8	2022-10-18	
6-SG-102-02	7.11	3.56	2016-02-12	5.43	5.10	0.085	0.090	17.2	2022-10-18	
6-SG-102-03	7.11	3.56	2016-02-12	5.22	4.92	0.090	0.082	15.1	2022-10-18	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-04	7.11	3.56	2016-02-12	5.18	4.90	0.089	0.076	15.1	2022-10-18	
6-SG-102-05	7.11	3.56	2016-02-12	6.19	6.07	0.045	0.033	>50	2022-10-18	
6-SG-102-06	7.11	3.56	2016-02-12	6.06	6.03	0.044	0.008	>50	2022-10-18	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2016-02-12	7.76	7.63	0.024	0.035	>50	2025-10-18	
8-TG-101-02	8.18	4.06	2016-02-12	7.46	7.33	0.037	0.035	>50	2025-10-18	
8-TG-101-03	8.18	4.06	2016-02-12	7.54	7.37	0.036	0.046	>50	2025-10-18	
8-TG-101-04	8.18	4.06	2016-02-12	7.51	7.38	0.030	0.035	>50	2025-10-18	
8-TG-101-05	8.18	4.06	2016-02-12	7.73	7.57	0.028	0.043	>50	2025-10-18	
8-TG-101-06	8.18	4.06	—	—	7.75	—	—	—	2025-10-18	
8-TG-101-07	8.18	4.06	2016-02-12	7.39	7.17	0.060	0.060	>50	2025-10-18	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2016-02-12	5.02	4.70	0.055	0.087	17.5	2025-10-18	
4-LA-101-02	6.02	3.18	2016-02-12	5.45	5.26	0.032	0.052	40.3	2025-10-18	
4-LA-101-03	6.02	3.18	2016-02-12	5.45	5.38	0.025	0.019	>50	2025-10-18	
4-LA-101-04	6.02	3.18	2016-02-12	5.48	5.29	0.033	0.052	40.9	2025-10-18	
4-LA-101-05	6.02	3.18	2016-02-12	5.54	5.38	0.027	0.043	>50	2025-10-18	
4-LA-101-06	6.02	3.18	2016-02-12	5.30	5.22	0.036	0.022	>50	2025-10-18	
4-LA-101-07	6.02	3.18	2016-02-12	4.87	4.86	0.048	0.003	35.1	2025-10-18	
4-LA-101-08	6.02	3.18	—	—	5.04	—	—	—	2025-10-18	
4-LA-101-09	6.02	3.18	2016-02-12	5.00	4.86	0.038	0.038	44.2	2025-10-18	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2016-02-12	5.25	5.09	0.039	0.043	43.9	2025-10-18	
4-LA-102-02	6.02	3.18	2016-02-12	5.36	5.15	0.039	0.057	34.5	2025-10-18	
4-LA-102-03	6.02	3.18	2016-02-12	5.28	5.15	0.039	0.035	>50	2025-10-18	
4-LA-102-04	6.02	3.18	2016-02-12	4.90	4.65	0.057	0.068	21.6	2025-10-18	
4-LA-102-05	6.02	3.18	2016-02-12	5.34	5.23	0.033	0.030	>50	2025-10-18	
4-LA-102-06	6.02	3.18	2016-02-12	5.60	5.52	0.020	0.022	>50	2025-10-18	
4-LA-102-07	6.02	3.18	2016-02-12	5.40	5.32	0.029	0.022	>50	2025-10-18	
4-LA-102-08	6.02	3.18	2016-02-12	5.70	5.55	0.041	0.041	>50	2025-10-18	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2016-02-12	6.44	6.40	0.026	0.011	>50	2025-10-18	
6-LA-103-02	7.11	3.56	2016-02-12	6.53	6.49	0.030	0.011	>50	2025-10-18	
6-LA-103-03	7.11	3.56	2016-02-12	6.28	6.23	0.036	0.014	>50	2025-10-18	
6-LA-103-04	7.11	3.56	2016-02-12	6.15	5.85	0.056	0.082	28.1	2025-10-18	
6-LA-103-05	7.11	3.56	2016-02-12	5.98	5.81	0.058	0.046	38.5	2025-10-18	
6-LA-103-06	7.11	3.56	2016-02-12	5.82	5.46	0.072	0.098	19.4	2025-10-18	
6-LA-103-07	7.11	3.56	2016-02-12	6.12	5.93	0.053	0.052	44.8	2025-10-18	
6-LA-103-08	7.11	3.56	2016-02-12	6.49	6.19	0.036	0.082	32.3	2025-10-18	
6-LA-103-09	7.11	3.56	—	—	6.16	—	—	—	2025-10-18	
6-LA-103-10	7.11	3.56	2016-02-12	6.30	6.00	0.082	0.082	29.9	2025-10-18	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2016-02-12	7.83	7.29	0.052	0.147	23.7	2022-10-18	
4-RA-101-02	8.56	3.81	2016-02-12	7.94	7.39	0.049	0.149	24.0	2022-10-18	
4-RA-101-03	8.56	3.81	2016-02-12	7.65	6.64	0.083	0.274	10.3	2022-10-18	
4-RA-101-04	8.56	3.81	2016-02-12	8.00	7.54	0.046	0.125	29.8	2022-10-18	
4-RA-101-05	8.56	3.81	2016-02-12	7.82	7.18	0.060	0.174	19.4	2022-10-18	
4-RA-101-06	8.56	3.81	2016-02-12	7.87	6.94	0.191	0.253	12.4	2022-10-18	
4-RA-101-07	8.56	3.81	2016-02-12	8.14	7.68	0.039	0.125	31.0	2022-10-18	
4-RA-101-08	8.56	3.81	2016-02-12	7.31	6.45	0.094	0.234	11.3	2022-10-18	
4-RA-101-09	8.56	3.81	2016-02-12	8.13	7.68	0.038	0.122	31.6	2022-10-18	
4-RA-101-10	8.56	3.81	2016-02-12	8.02	7.55	0.047	0.128	29.3	2022-10-18	
4-RA-101-11	8.56	3.81	2016-02-12	7.60	6.66	0.086	0.255	11.2	2022-10-18	
4-RA-101-12	8.56	3.81	—	—	7.03	—	—	—	2022-10-18	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2016-02-12	8.02	7.58	0.041	0.120	31.5	2022-10-18	
4-RA-102-02	8.56	3.81	2016-02-12	7.64	6.96	0.068	0.185	17.0	2022-10-18	
4-RA-102-03	8.56	3.81	2016-02-12	7.55	7.05	0.064	0.136	23.8	2022-10-18	
4-RA-102-04	8.56	3.81	2016-02-12	7.72	7.04	0.064	0.185	17.5	2022-10-18	
4-RA-102-05	8.56	3.81	2016-02-12	7.67	7.06	0.063	0.166	19.6	2022-10-18	
4-RA-102-06	8.56	3.81	2016-02-12	7.73	7.15	0.058	0.158	21.2	2022-10-18	
4-RA-102-07	8.56	3.81	2016-02-12	7.78	7.30	0.054	0.130	26.8	2022-10-18	
4-RA-102-08	8.56	3.81	2016-02-12	7.35	6.40	0.089	0.258	10.0	2022-10-18	
4-RA-102-09	8.56	3.81	2016-02-12	7.54	6.78	0.075	0.207	14.4	2022-10-18	
4-RA-102-10	8.56	3.81	—	—	7.36	—	—	—	2022-10-18	
4-RA-102-11	8.56	3.81	2016-02-12	7.96	7.72	0.054	0.065	>50	2022-10-18	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2016-02-12	10.29	9.83	0.050	0.125	42.1	2022-10-18	
6-RA-103-02	10.97	4.57	2016-02-12	9.69	8.61	0.105	0.294	13.8	2022-10-18	
6-RA-103-03	10.97	4.57	2016-02-12	10.06	9.49	0.064	0.155	31.8	2022-10-18	
6-RA-103-04	10.97	4.57	2016-02-12	9.31	8.22	0.117	0.296	12.3	2022-10-18	
6-RA-103-05	10.97	4.57	2016-02-12	9.48	8.33	0.111	0.313	12.0	2022-10-18	
6-RA-103-06	10.97	4.57	2016-02-12	10.32	9.79	0.052	0.144	36.2	2022-10-18	
6-RA-103-07	10.97	4.57	2016-02-12	10.20	9.59	0.059	0.166	30.3	2022-10-18	
6-RA-103-08	10.97	4.57	2016-02-12	10.33	9.73	0.053	0.163	31.6	2022-10-18	
6-RA-103-09	10.97	4.57	2016-02-12	10.33	9.78	0.052	0.149	34.9	2022-10-18	
6-RA-103-10	10.97	4.57	2016-02-12	9.49	8.37	0.112	0.304	12.5	2022-10-18	
6-RA-103-11	10.97	4.57	2016-02-12	9.77	8.91	0.234	0.234	18.5	2022-10-18	
6-RA-103-12	10.97	4.57	—	—	9.52	—	—	—	2022-10-18	
6-RA-103-13	10.97	4.57	2016-02-12	10.15	9.55	0.104	0.163	30.5	2022-10-18	
6-RA-103-14	10.97	4.57	—	—	8.26	—	—	—	2022-10-18	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2016-02-12	6.34	6.05	0.065	0.079	31.6	2022-10-18	
3-RV-101-02	7.62	3.56	2016-02-12	6.31	5.97	0.068	0.092	26.1	2022-10-18	
3-RV-101-03	7.62	3.56	2016-02-12	5.63	5.24	0.097	0.106	15.9	2022-10-18	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-04	7.62	3.56	2016-02-12	5.88	5.58	0.085	0.082	23.7	2022-10-18	
3-RV-101-05	7.62	3.56	2016-02-12	6.36	6.11	0.066	0.068	37.5	2022-10-18	
3-RV-101-06	7.62	3.56	2016-02-12	5.96	5.55	0.086	0.111	17.9	2022-10-18	
3-RV-101-07	7.62	3.56	2016-02-12	5.54	5.07	0.124	0.128	11.8	2022-10-18	
3-RV-101-08	7.62	3.56	2016-02-12	6.31	5.86	0.093	0.122	18.8	2022-10-18	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2016-02-12	4.35	4.16	0.058	0.052	23.6	2022-10-18	
2-RV-102-02	5.54	2.79	2016-02-12	4.01	3.86	0.073	0.041	14.6	2022-10-18	
2-RV-102-03	5.54	2.79	2016-02-12	4.64	4.37	0.047	0.073	21.5	2022-10-18	
2-RV-102-04	5.54	2.79	2016-02-12	4.73	4.59	0.042	0.038	42.9	2022-10-18	
2-RV-102-05	5.54	2.79	2016-02-12	4.97	4.78	0.029	0.052	38.5	2022-10-18	
2-RV-102-06	5.54	2.79	2016-02-12	4.27	4.27	0.060	0.000	24.5	2022-10-18	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2016-02-12	5.31	5.40	0.003	-0.024	>50	2025-10-18	
3-RF-101-02	5.49	2.41	2016-02-12	5.34	5.38	0.003	-0.011	>50	2025-10-18	
3-RF-101-03	5.49	2.41	2016-02-12	5.33	5.35	0.004	-0.005	>50	2025-10-18	
3-RF-101-04	5.49	2.41	2016-02-12	5.35	5.29	0.007	0.016	>50	2025-10-18	
3-RF-101-05	5.49	2.41	2016-02-12	5.33	5.17	0.012	0.043	>50	2025-10-18	
3-RF-101-06	5.49	2.41	2016-02-12	5.23	5.12	0.017	0.030	>50	2025-10-18	
3-RF-101-07	5.49	2.41	2016-02-12	5.37	5.38	0.002	-0.003	>50	2025-10-18	
3-RF-101-08	5.49	2.41	—	—	5.23	—	—	—	2025-10-18	
3-RF-101-09	5.49	2.41	2016-02-12	5.19	5.13	-0.002	0.016	>50	2025-10-18	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2016-02-12	3.76	3.66	0.011	0.027	>50	2025-10-18	
2-RF-102-02	3.91	2.16	2016-02-12	3.64	3.65	0.010	-0.003	>50	2025-10-18	
2-RF-102-03	3.91	2.16	2016-02-12	3.75	3.68	0.011	0.019	>50	2025-10-18	
2-RF-102-04	3.91	2.16	2016-02-12	3.38	2.83	0.044	0.149	4.5	2022-01-14	RL/2 < interval
2-RF-102-05	3.91	2.16	2016-02-12	3.49	2.88	0.133	0.166	4.3	2021-12-19	RL/2 < interval
2-RF-102-06	3.91	2.16	2016-02-12	3.47	2.74	0.159	0.198	2.9	2021-04-04	RL/2 < interval
2-RF-102-07	3.91	2.16	2016-02-12	3.57	2.82	0.048	0.204	3.2	2021-06-01	RL/2 < interval

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2016-02-12	5.31	5.16	0.039	0.041	48.6	2025-10-18	
4-LA-104-02	6.02	3.18	2016-02-12	5.00	4.79	0.053	0.057	28.2	2025-10-18	
4-LA-104-03	6.02	3.18	2016-02-12	5.42	5.23	0.031	0.052	39.7	2025-10-18	
4-LA-104-04	6.02	3.18	2016-02-12	5.63	5.49	0.024	0.038	>50	2025-10-18	
4-LA-104-05	6.02	3.18	2016-02-12	4.88	4.71	0.056	0.046	27.2	2025-10-18	
4-LA-104-06	6.02	3.18	2016-02-12	5.55	5.38	0.030	0.046	47.6	2025-10-18	
4-LA-104-07	6.02	3.18	2016-02-12	5.28	5.15	0.036	0.035	>50	2025-10-18	
4-LA-104-08	6.02	3.18	2016-02-12	5.11	4.88	0.045	0.063	27.2	2025-10-18	
4-LA-104-09	6.02	3.18	—	—	4.61	—	—	—	2025-10-18	
4-LA-104-10	6.02	3.18	2016-02-12	4.84	4.63	0.057	0.057	25.4	2025-10-18	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2016-02-12	7.71	7.15	0.058	0.152	21.9	2022-10-18	
4-RA-104-02	8.56	3.81	2016-02-12	7.85	7.31	0.053	0.147	23.8	2022-10-18	
4-RA-104-03	8.56	3.81	2016-02-12	8.28	7.94	0.071	0.092	44.7	2022-10-18	
4-RA-104-04	8.56	3.81	2016-02-12	8.18	7.64	0.097	0.147	26.1	2022-10-18	
4-RA-104-05	8.56	3.81	2016-02-12	8.16	7.57	0.095	0.160	23.5	2022-10-18	
4-RA-104-06	8.56	3.81	2016-02-12	7.61	7.13	0.062	0.130	25.5	2022-10-18	
4-RA-104-07	8.56	3.81	2016-02-12	7.32	6.67	0.081	0.177	16.2	2022-10-18	
4-RA-104-08	8.56	3.81	2016-02-12	7.81	7.46	0.083	0.095	38.4	2022-10-18	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2016-02-12	3.61	3.59	0.014	0.005	>50	2025-10-18	
2-ST-101-02	3.91	2.29	2016-02-12	3.55	3.34	0.024	0.057	18.4	2025-10-18	
2-ST-101-03	3.91	2.29	2016-02-12	3.55	3.68	0.011	-0.035	>50	2025-10-18	
2-ST-101-04	3.91	2.29	2016-02-12	3.68	3.52	0.013	0.043	28.3	2025-10-18	
2-ST-101-05	3.91	2.29	2016-02-12	3.50	3.27	0.027	0.063	15.7	2025-10-18	
2-ST-101-06	3.91	2.29	2016-02-12	3.63	3.66	-0.008	-0.008	>50	2025-10-18	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2016-02-12	5.03	4.91	0.022	0.033	>50	2025-10-18	
3-CW-101-02	5.49	2.79	2016-02-12	5.01	4.98	0.018	0.008	>50	2025-10-18	
3-CW-101-03	5.49	2.79	2016-02-12	4.93	4.87	0.028	0.016	>50	2025-10-18	
3-CW-101-04	5.49	2.79	2016-02-12	5.26	5.20	0.011	0.016	>50	2025-10-18	
3-CW-101-05	5.49	2.79	2016-02-12	5.35	5.14	0.014	0.057	41.2	2025-10-18	
3-CW-101-06	5.49	2.79	2016-02-12	5.08	5.01	0.019	0.019	>50	2025-10-18	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2016-02-12	5.06	4.85	0.028	0.057	36.1	2025-10-18	
3-CW-102-02	5.49	2.79	2016-02-12	4.37	4.17	0.057	0.054	24.3	2025-10-18	
3-CW-102-03	5.49	2.79	2016-02-12	4.54	4.27	0.053	0.073	20.2	2025-10-18	
3-CW-102-04	5.49	2.79	2016-02-12	4.69	4.50	0.043	0.052	33.1	2025-10-18	
3-CW-102-05	5.49	2.79	2016-02-12	4.66	4.45	0.057	0.057	29.1	2025-10-18	

Sign-off	Name	Function	Date	Signature
Prepared	Trevor Nolan	UT technician, SNT-TC-1A UT Level II	2019-10-18	_____